

Centralized Audit Trail Compliance Engine

NAME: SAGAR K

SRN: PES2UG24CS424

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall parse structured JSON audit logs and identify and mask sensitive PII fields such as Social Security Numbers and credit card numbers before persisting records to storage.	High	Stored audit logs contain masked PII and a SHA-256 integrity hash.	Protects sensitive personal information and supports GDPR compliance.	Given requirement
FR-002	Functional	The system shall receive and validate audit logs submitted by enterprise applications.	High	A valid JSON audit log is accepted and processed.	Ensures that only valid audit events enter the compliance system.	Validation should reject malformed JSON.
FR-003	Functional	The system shall generate and store a SHA-256 integrity hash for every processed audit log.	High	Every persisted audit record contains a SHA-256 hash and recomputing the hash produces the same value.	Provides cryptographic evidence that audit records have not been modified.	Hash must be generated after PII masking.
FR-004	Functional	The system shall allow a Compliance Officer to search and filter audit logs using fields such as date, user, event type, and severity.	High	A valid filter returns all matching audit records without exposing unmasked PII.	Enables compliance officers to investigate and review audit activity efficiently.	Filters should support multiple conditions.
FR-005	Functional	The system shall allow a Security Auditor to export selected audit records as a cryptographically verifiable immutable audit report.	High	The exported report contains the selected masked records and integrity hashes and can be successfully verified.	Supports audits and provides evident compliance evidence	Export should preserve integrity information.
FR-006	Non - Functional	The system shall support complex filtering across 10 million historical audit records in under 1 second.	High	Benchmarking tests shows query response time less than 1 sec under the defined workload	Ensure fast access to historical audit information during compliance officer investigation.	Given requirements.
FR-007	Non - Functional	The system must protect the audit records and exported records from unauthorized access.	High	The unauthorised users cannot modify the audit records. Modification attempts are detected from integrity verification.	Audit records contain sensitive and legal information.	Access control and integrity verification must be enforced.

Actors

1. Application / Enterprise System

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2. Compliance Officer
3. Security Auditor

Main Use Cases

1. Submit Audit Log
2. Parse and Validate Audit Log
3. Mask PII
4. Store Audit Log
5. Generate Integrity Hash
6. Search and Filter Audit Logs
7. Export Immutable Audit Report
8. Verify Audit Integrity

Important UML Relationships

- Submit Audit Log «include» Parse and Validate Audit Log
- Parse and Validate Audit Log «include» Mask PII
- Store Audit Log «include» Generate Integrity Hash
- Export Immutable Audit Report «include» Verify Audit Integrity.